

Research Statement Outline

Temporal Dynamics and Domain Generalization in Emotion Recognition for ASD Support

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1. Introduction and Vision

Automated understanding of social interactions in clinical settings remains a major challenge due to the subtlety of emotional expressions and the variability of real-world environments. My research vision focuses on developing **Robust Video Representation Learning** that captures both the spatial details of micro-expressions and the temporal dynamics of social synchrony. By leveraging **State Space Models (SSMs)** and **Vision Transformers**, I aim to build a diagnostic support system that is resilient to the "distribution shift" between generic movie datasets and naturally occurring medical interactions.

2. Proposed Research Axes

2.1. 1. Modeling Emotional Synchrony via State Space Models (SSMs)

Traditional RNNs and LSTMs often struggle with very long video sequences. I propose to investigate the use of **Mamba-SSM** architectures to capture the long-range temporal dependencies inherent in social interactions. This will focus on "emotional synchrony"—the rhythmic coordination between a child and a caregiver. By modeling the interaction as a continuous state-space, we can better identify the subtle shifts in behavioral patterns that are critical for ASD assessment but often missed by frame-level analysis.

2.2. 2. Fine-grained FER for Low-Intensity Expressions

Standard Facial Expression Recognition (FER) models are often over-fitted to exaggerated emotions. I propose to adapt **Multimodal Foundation Models** (such as CLIP or Video-MAV) using specialized datasets of subtle, day-to-day expressions. This involves developing custom attention mechanisms that prioritize small facial muscle movements (Action Units) over global facial structure, ensuring the model remains sensitive even when the visual signal-to-noise ratio is low due to distance or suboptimal camera placement.

2.3. 3. Domain Generalization via Latent Space Alignment

Deployment in unconstrained scenes requires models that are robust to illumination and pose variations. I propose to investigate **Domain Generalization** techniques that align the latent representations of clinical data with those of large-scale generic datasets. By using self-supervised pre-training and contrastive learning, we can force the model to learn "invariant" emotion features that persist regardless of the background or the specific camera viewpoint.

3. Alignment with AIMAIN and LIRIS

My background as an AI Engineer with expertise in **Real-time Computer Vision** (Recognizini) and **Advanced Sequential Modeling** (Sonatrach Thesis) provides a unique foundation for this PhD. I have demonstrated the ability to bridge complex mathematical theory with practical, high-performance implementation. I am eager to collaborate with the interdisciplinary

team at LIRIS to translate state-of-the-art AI into a meaningful tool for ASD diagnosis and healthcare support.